

H524 Introduction to Biostatistics

Week 2: Probability Concepts and Applications

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Outline

- 1 Review from Week 1
- 2 Basic Probability Concepts (Ch 6.1)
- 3 Conditional Probability (Ch 6.2)
- 4 Diagnostic Tests (Ch 6.3)
- 5 Random Variables and Expected Value (Ch 6.4)
- 6 Discrete Distributions (Ch 6.5)
- 7 Applications and Examples
- 8 AI as a Learning Tool for Probability
- 9 R Examples and Exercises

Quick Review: Key Concepts

- **Data Types:** Quantitative vs Qualitative
- **Data Presentation:** Histograms, box plots, bar charts
- **Summary Measures:** Central tendency and variability
- **Rates and Standardization:** Age-specific rates, direct/indirect methods

This Week: Chapter 6 - Probability foundations for statistical inference

Reading: Pagano & Gauvreau Chapter 6 (6.1-6.4.2, 6.5)

Why Probability in Biostatistics?

- **Foundation for Statistical Inference**

- Understanding uncertainty in medical research
- Bridge between sample data and population conclusions

- **Clinical Applications**

- Diagnostic test interpretation
- Treatment efficacy assessment
- Risk communication to patients

- **Research Design**

- Power calculations
- Sample size determination
- Study planning and interpretation

Basic Probability Concepts

- **Experiment:** Any process that generates observations
 - Example: Testing a patient for COVID-19
- **Sample Space (S):** Set of all possible outcomes
 - Example: $S = \{\text{Positive}, \text{Negative}\}$
- **Event:** Subset of the sample space
 - Example: Patient tests positive
- **Probability:** Measure of likelihood $[0, 1]$
 - $\Pr(\text{Event}) = \frac{\text{Number of favorable outcomes}}{\text{Total number of possible outcomes}}$

- ➊ **Range:** $0 \leq \Pr(A) \leq 1$ for any event A
- ➋ **Certainty:** $\Pr(S) = 1$ (something must happen)
- ➌ **Complement:** $\Pr(A^c) = 1 - \Pr(A)$
- ➍ **Addition Rule:**

$$\Pr(A \cup B) = \Pr(A) + \Pr(B) - \Pr(A \cap B)$$

- ➎ **Multiplication Rule:**

$$\Pr(A \cap B) = \Pr(A) \times \Pr(B|A)$$

COVID-19 Rapid Test Performance:

- Sensitivity = 85% (probability of positive test given disease)
- Specificity = 97% (probability of negative test given no disease)
- Disease prevalence = 5%

Questions:

- What's the probability of a false positive?
- What's the probability someone with a positive test has COVID?
- How does prevalence affect test interpretation?

These questions require understanding conditional probability!

Conditional Probability

Definition: Probability of event A given that event B has occurred

$$\Pr(A|B) = \frac{\Pr(A \cap B)}{\Pr(B)}, \text{ where } \Pr(B) > 0$$

Medical Example:

- A = Patient has diabetes
- B = Patient is overweight
- $\Pr(A|B)$ = Probability of diabetes given patient is overweight

Key Insight: Conditional probability updates our beliefs based on new information

Definition: Events A and B are independent if:

$$\Pr(A|B) = \Pr(A) \text{ or equivalently } \Pr(A \cap B) = \Pr(A) \times \Pr(B)$$

Examples in Healthcare:

- **Independent:**

- Blood type and eye color
- Two different patients' test results

- **Dependent:**

- Smoking status and lung cancer
- Age and cardiovascular disease
- Family history and genetic disorders

Caution: Independence is often assumed but should be verified!

Key Metrics for Medical Tests:

- **Sensitivity:** $\Pr(\text{Positive Test} \mid \text{Disease Present})$
 - Ability to correctly identify disease
 - True positive rate
- **Specificity:** $\Pr(\text{Negative Test} \mid \text{Disease Absent})$
 - Ability to correctly identify no disease
 - True negative rate

Example - COVID-19 Rapid Test:

- Sensitivity = 85% (detects 85% of infected people)
- Specificity = 97% (correctly identifies 97% of non-infected)

Clinical Interpretation Measures:

- **Positive Predictive Value (PPV):** $\Pr(\text{Disease} \mid \text{Positive Test})$
 - Probability patient has disease given positive test
 - Depends on disease prevalence
- **Negative Predictive Value (NPV):** $\Pr(\text{No Disease} \mid \text{Negative Test})$
 - Probability patient is healthy given negative test
 - Also depends on disease prevalence

Key Insight: PPV and NPV change with disease prevalence in the population, while sensitivity and specificity are test characteristics.

A Counterintuitive Example

HIV Screening Test with Excellent Performance:

- Sensitivity = 99.5% (detects 995 of 1000 infected people)
- Specificity = 98.5% (correctly identifies 985 of 1000 uninfected)
- Population prevalence = 0.1% (1 in 1000 people)

Question: If you test positive, what's the probability you have HIV?

A Counterintuitive Example

HIV Screening Test with Excellent Performance:

- Sensitivity = 99.5% (detects 995 of 1000 infected people)
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- Population prevalence = 0.1% (1 in 1000 people)

Question: If you test positive, what's the probability you have HIV?

Approach: Consider 100,000 people tested

	HIV +	HIV -	Total
Test +	100	1,498	1,598
Test -	0	98,402	98,402
Total	100	99,900	100,000

$$\text{PPV} = \frac{100}{1,598} = 0.063 \text{ or } \mathbf{6.3\%}$$

Surprising Result: Only 6.3% of positive tests indicate actual infection!

Why? With low prevalence, false positives (1,498) vastly outnumber true positives (100).

Understanding Odds

Probability vs Odds:

- **Probability:** $\Pr(\text{Event}) = \frac{\text{Number with event}}{\text{Total number}}$
- **Odds:** $\text{Odds}(\text{Event}) = \frac{\text{Number with event}}{\text{Number without event}}$

Relationship: $\text{Odds} = \frac{p}{1-p}$ where p is the probability

Example: Risk of lung cancer in smokers

- 60 smokers develop lung cancer, 440 don't
- **Risk (probability):** $\frac{60}{500} = 0.12$ (12%)
- **Odds:** $\frac{60}{440} = 0.136$ (or "60 to 440" or "3 to 22")

Key insight: When probability is small (< 0.10), odds $\approx$ probability

- **Relative Risk (RR):**

- Direct measure of risk comparison
- Used in cohort studies
- $RR = 2$ means exposed have twice the risk
- Cannot be calculated in case-control studies

- **Odds Ratio (OR):**

- Approximates RR when disease is rare
- Can be calculated in all study designs
- More difficult to interpret directly
- $OR > RR$ when outcome is common

Rule of thumb: When disease prevalence $< 10\%$, $OR \approx RR$

Example: Smoking and Lung Cancer

Study of 1000 individuals over 20 years:

	Lung Cancer	No Cancer	Total
Smokers	60	440	500
Non-smokers	10	490	500
Total	70	930	1000

Calculations:

- Risk in smokers: $60/500 = 0.12$ (12%)
- Risk in non-smokers: $10/500 = 0.02$ (2%)
- Relative Risk: $0.12/0.02 = 6.0$
- Odds Ratio: $(60 \times 490)/(10 \times 440) = 6.68$

Interpretation: Smokers have 6 times the risk of lung cancer

Definition: Long-run average value of a random variable

Discrete Random Variable:

$$E(X) = \mu = \sum_i x_i \cdot \Pr(X = x_i)$$

Medical Example: Number of adverse events in clinical trial

Events (x)	0	1	2	3
Probability	0.70	0.20	0.08	0.02

$$E(X) = 0(0.70) + 1(0.20) + 2(0.08) + 3(0.02) = 0.42$$

Interpretation: On average, expect 0.42 adverse events per patient

Variance and Standard Deviation

Variance: Measure of spread around the expected value

$$\text{Var}(X) = \sigma^2 = E[(X - \mu)^2] = E(X^2) - [E(X)]^2$$

Standard Deviation: Square root of variance

$$SD(X) = \sigma = \sqrt{\text{Var}(X)}$$

Properties:

- $\text{Var}(aX) = a^2\text{Var}(X)$
- $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ if X and Y are independent
- Standard deviation has same units as original data

Binomial Distribution

Conditions:

- Fixed number of trials (n)
- Two possible outcomes (success/failure)
- Constant probability of success (p)
- Independent trials

Probability Mass Function:

$$\Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Parameters:

- Mean: $\mu = np$
- Variance: $\sigma^2 = np(1 - p)$

Binomial Example: Vaccine Efficacy

Clinical Trial: 20 vaccinated individuals exposed to virus

- Vaccine efficacy = 90% (probability of protection)
- What's the probability exactly 18 are protected?

Solution:

- $n = 20, p = 0.90, k = 18$
- $\Pr(X = 18) = \binom{20}{18}(0.90)^{18}(0.10)^2$
- $\Pr(X = 18) = 190 \times 0.1501 \times 0.01 = 0.285$

Expected number protected: $\mu = 20 \times 0.90 = 18$

Standard deviation: $\sigma = \sqrt{20 \times 0.90 \times 0.10} = 1.34$

Normal Distribution

Properties:

- Bell-shaped, symmetric curve
- Characterized by mean (μ) and standard deviation (σ)
- Area under curve = 1
- Approximately 68% within $\pm 1\sigma$
- Approximately 95% within $\pm 2\sigma$
- Approximately 99.7% within $\pm 3\sigma$

Standard Normal: $Z = \frac{X - \mu}{\sigma}$ has mean 0, SD 1

Medical Applications:

- Blood pressure measurements
- Cholesterol levels
- Height and weight
- Many lab values

Normal Distribution Example

Systolic Blood Pressure in Adults:

- Mean = 120 mmHg
- Standard deviation = 15 mmHg
- What proportion have BP > 140 mmHg (hypertension)?

Solution:

- 1 Standardize: $Z = \frac{140-120}{15} = 1.33$
- 2 Look up in Z-table: $\Pr(Z < 1.33) = 0.9082$
- 3 Therefore: $\Pr(BP > 140) = 1 - 0.9082 = 0.0918$

Interpretation: About 9.2% of adults have hypertension by this criterion

Disease Surveillance Example:

- Population: 50,000 individuals
- New disease cases: 125 per year
- Deaths from disease: 15 per year

Calculate probabilities:

- $\Pr(\text{disease}) = 125/50,000 = 0.0025$ (0.25%)
- $\Pr(\text{death} \mid \text{disease}) = 15/125 = 0.12$ (12%)
- $\Pr(\text{death and disease}) = 15/50,000 = 0.0003$ (0.03%)

Clinical interpretation:

- Incidence rate: 250 per 100,000 per year
- Case fatality rate: 12%
- Overall mortality rate: 30 per 100,000 per year

How AI can help with probability:

- **Complex calculations:** Conditional probabilities, combinatorics
- **Simulation:** Generate random data, bootstrap samples
- **Visualization:** Create probability distributions
- **Problem solving:** Step-by-step solutions
- **Code generation:** R scripts for probability problems

Example prompts:

- “Calculate the positive predictive value for a test with 95% sensitivity, 90% specificity, and 5% prevalence”
- “Simulate 10,000 binomial trials with $n=20$, $p=0.3$ in R”
- “Explain the difference between relative risk and odds ratio with a medical example”

AI Demo: Diagnostic Test Calculations

Prompt: "Create R code to calculate diagnostic test metrics"

```
# Diagnostic Test Calculator
diagnostic_metrics <- function(sens, spec, prev) {
  # Calculate PPV and NPV
  ppv <- (sens * prev) /
    (sens * prev + (1-spec) * (1-prev))
  npv <- (spec * (1-prev)) /
    ((1-sens) * prev + spec * (1-prev))

  # Display results
  cat("PPV:", round(ppv*100, 1), "%\n")
  cat("NPV:", round(npv*100, 1), "%\n")

  return(list(PPV=ppv, NPV=npv))
}

# Example: COVID test
diagnostic_metrics(sens=0.85, spec=0.97, prev=0.05)
```

AI Demo: Probability Simulations

Prompt: "Simulate coin flips to estimate probability"

```
# Simulate binomial trials
simulate_treatment <- function(n_patients=100,
                               p_success=0.7) {
  outcomes <- rbinom(n_patients, 1, p_success)
  success_rate <- mean(outcomes)

  cat("Simulated success rate:",
      round(success_rate*100, 1), "%\n")
  cat("Expected success rate:", p_success*100, "%\n")

  # Create visualization
  barplot(table(outcomes),
          names.arg=c("Failure", "Success"),
          col=c("red", "green"),
          main="Treatment Outcomes")
}

simulate_treatment()
```

AI Demo: Visualizing Distributions

Prompt: “Compare binomial and normal distributions in R”

```
# Compare Binomial to Normal
n <- 100; p <- 0.3
x <- 0:n

# Binomial probabilities
binom_probs <- dbinom(x, n, p)

# Normal approximation
mu <- n * p
sigma <- sqrt(n * p * (1-p))

# Plot comparison
plot(x, binom_probs, type="h", col="blue",
     main="Binomial vs Normal",
     xlab="x", ylab="Probability")
curve(dnorm(x, mu, sigma), add=TRUE, col="red")
legend("topright", c("Binomial", "Normal"),
     col=c("blue", "red"), lty=1)
```

AI Best Practices for Probability

DO:

- Verify calculations manually
- Check assumptions are met
- Request step-by-step solutions
- Ask for multiple approaches
- Validate with simulations
- Cross-reference formulas

DON'T:

- Accept complex calculations blindly
- Skip understanding the logic
- Ignore edge cases
- Misinterpret conditional probability
- Confuse probability types
- Trust without verification

Remember: AI can calculate quickly, but you need to understand the concepts!

AI Task: Probability Problem Solving

This week's AI-integrated tasks:

- 1 Use AI to solve a diagnostic test problem, then verify manually
- 2 Generate R code for simulating coin flips, then modify for medical scenario
- 3 Ask AI to explain conditional probability three different ways
- 4 Create visualizations of different probability distributions
- 5 Identify and correct an AI error in probability calculation

Sample Problem:

“A new diagnostic test for Disease X has 92% sensitivity and 88% specificity. In a population where 3% have the disease, what is the probability that someone with a positive test actually has the disease? Create R code to calculate this and visualize how the answer changes as prevalence varies from 1% to 10%.”

R Example: Basic Probability

```
# Probability calculations in R
# Sample space for rolling two dice
dice1 <- 1:6
dice2 <- 1:6
sample_space <- expand.grid(dice1, dice2)
sample_space$sum <- sample_space$Var1 + sample_space$Var2

# Probability of sum = 7
p_seven <- sum(sample_space$sum == 7) / nrow(sample_space)
cat("Pr(sum = 7) =", p_seven, "\n") # 6/36 = 1/6

# Conditional probability example
# Pr(sum > 8 | first die = 4)
subset_data <- sample_space[sample_space$Var1 == 4, ]
p_conditional <- sum(subset_data$sum > 8) / nrow(subset_data)
cat("Pr(sum > 8 | first = 4) =", p_conditional, "\n") # 3/6 = 1/2
```

R Example: Medical Testing

```
# Breast cancer screening example
sens <- 0.90 # Sensitivity
spec <- 0.95 # Specificity
prev <- 0.008 # Prevalence

# Calculate PPV and NPV
ppv <- (sens * prev) /
      (sens * prev + (1-spec) * (1-prev))
npv <- (spec * (1-prev)) /
      ((1-sens) * prev + spec * (1-prev))

cat("PPV:", round(ppv * 100, 1), "%\n")
cat("NPV:", round(npv * 100, 1), "%\n")

# Simulate population
n <- 10000
n_disease <- round(n * prev)
n_healthy <- n - n_disease

# Test outcomes
TP <- round(n_disease * sens)
```

R Example: Binomial Distribution

```
# Clinical trial with 20 patients, treatment success rate 70%
n <- 20
p <- 0.7

# Probability of exactly 15 successes
p_15 <- dbinom(15, n, p)
cat("Pr(X = 15) =", round(p_15, 4), "\n")

# Probability of at least 15 successes
p_atleast_15 <- pbinom(14, n, p, lower.tail = FALSE)
cat("Pr(X >= 15) =", round(p_atleast_15, 4), "\n")

# Expected value and standard deviation
expected <- n * p
std_dev <- sqrt(n * p * (1-p))
cat("Expected successes:", expected, "\n")
cat("Standard deviation:", round(std_dev, 2), "\n")

# Visualize distribution
x <- 0:n
probs <- dbinom(x, n, p)
```

R Example: Normal Distribution

```
# Blood pressure in adults
mu <- 120      # Mean SBP
sigma <- 15    # Standard deviation

# Proportion with hypertension (SBP > 140)
p_hyper <- pnorm(140, mu, sigma, lower.tail = FALSE)
cat("Pr(BP > 140):", round(p_hyper * 100, 1), "%\n")

# Find percentiles
bp_90 <- qnorm(0.90, mu, sigma)
bp_95 <- qnorm(0.95, mu, sigma)
cat("90th percentile:", round(bp_90, 1), "mmHg\n")
cat("95th percentile:", round(bp_95, 1), "mmHg\n")

# Visualize distribution
x <- seq(70, 170, length = 100)
plot(x, dnorm(x, mu, sigma), type = "l",
     main = "Systolic Blood Pressure",
     xlab = "SBP (mmHg)", ylab = "Density")
abline(v = c(mu, 140), col = c("blue", "red"))
```

Summary: Chapter 6 - Probability

Today we covered:

- **Section 6.1:** Basic probability concepts, rules, and definitions
- **Section 6.2:** Conditional probability and independence
- **Section 6.3:** Diagnostic test applications (sensitivity, specificity)
- **Section 6.4:** Random variables and expected value
- **Section 6.5:** Discrete distributions (binomial)

Key Learning Objectives:

- Calculate and interpret basic probabilities
- Understand conditional probability and independence
- Apply probability concepts to diagnostic testing
- Use probability distributions for modeling

Next Week: Chapter 7 & 8 - Theoretical distributions and sampling

Thank you!

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Reading for next week:

Pagano & Gauvreau, Chapter 7 (7.1, 7.2, 7.4) and
Chapter 8